Instacart Customer Retention & Order Cadence Analysis

A cohort/retention analysis of Instacart order data | SQL (PostgreSQL) · Python · Power BI

Executive Summary

This analysis examines repeat-purchase behavior across a 25,000-user sample from the Instacart Market Basket dataset (~410,000 orders). Because Instacart/Kaggle pre-filtered this public dataset to include only customers with between 4 and 100 historical orders, the analysis focuses on **repeat-customer drop-off and behavior** rather than new-customer conversion. Four questions were examined: order-sequence retention, the effect of first-order product category, basket size stability, and order cadence (days between orders) by engagement level.

5.5%	54%	~10 items	19 vs 8 days
of customers reach order #50, down from 96% at order #4	repeat rate (10+ orders) for produce/dairy-first customers vs. 44-48% for shelf-stable-first customers	average basket size — stable across a customer's entire order history	average days between orders: low-engagement (4-6 orders) vs. highly-retained (20+ orders) customers

Methodology

Source: Instacart Market Basket Analysis dataset (public Kaggle release; ~3M orders, ~200K users). **Sample:** 25,000 users selected at the user level (not row-level), preserving each sampled user's complete order history — required for valid sequence-based cohort analysis. Only orders with product-level detail (eval_set = prior/train) were retained. **Tools:** PostgreSQL for cohort table construction, Python/pandas for data loading, cleaning and validation, Power BI for the interactive dashboard.

Key data caveat: Kaggle's published version of this dataset only includes customers who already placed between 4 and 100 orders — true one-time or two-time customers are not present in the data. This means questions originally framed around "1st vs. 2nd order retention" and "one-time vs. repeat customers" were reframed around this dataset's real floor: retention is tracked from order #4 onward, and the "low-engagement" vs. "highly-retained" comparison uses order-count thresholds (4-6 orders vs. 20+ orders) instead of a literal one-time/repeat split. This is disclosed here and on the dashboard's summary page.

Findings

1. Retention decays steadily after the dataset's order-4 floor

Among the 25,000 sampled users (100% of whom placed at least 4 orders, per the dataset's own inclusion rule), retention falls from 95.9% at order 4 to 85.1% at order 5, 76.2% at order 6, 52.5% at order 10, 26.0% at order 20, and just 5.5% at order 50. The single sharpest drop occurs between orders 4 and 5 — a ~11-point decline right at the dataset's inclusion threshold.

2. First-order category correlates with long-term engagement

Customers whose first order was dominated by perishable, restock-driven categories placed more total orders on average and were more likely to become highly engaged (10+ orders): produce-first customers averaged 17.3 orders with a 54.2% repeat rate, dairy/eggs-first customers averaged 17.3 orders with a 54.0% repeat rate, and babies-first customers (n=246, smaller sample) averaged 19.6 orders with a 60.6% repeat rate. By contrast, customers whose first order centered on shelf-stable or occasional categories showed consistently lower engagement: canned goods (13.1 avg orders, 43.5% repeat), personal care (12.4 avg orders, 47.6% repeat), and pantry (14.0 avg orders, 47.8% repeat).

3. Basket size is stable across a customer's entire order history

Average basket size stays within a narrow 9.9-10.4 item range from order 2 through order 30, with no meaningful growth or decline trend. This is a useful contrast to findings 1 and 2: while whether a customer returns varies substantially by category and time, how much they buy per trip is essentially constant once they're an established shopper.

4. Order cadence is the clearest behavioral differentiator between engagement levels

Low-engagement customers (4-6 total orders) had an average gap of 19.28 days between orders (std. dev. 10.46 — irregular). Highly-retained customers (20+ total orders) had an average gap of just 7.80 days (std. dev. 6.67 — a tight, near-weekly routine). This is the strongest single signal in the analysis: customers who settle into a roughly weekly cadence are the ones who stay long-term, while customers with large, irregular gaps tend to disengage near the dataset's order-4-6 floor.

Business Recommendations

1. Trigger re-engagement outreach on cadence, not just recency

Since low-engagement customers average 19+ days between orders (vs. 8 for retained customers), a customer who passes ~14 days without ordering while still early in their lifecycle (under 10 total orders) is a high-risk signal. Recommend a lifecycle trigger (reminder push/email or targeted promo) at the 14-day mark for customers in this segment, rather than a flat recency rule applied to all customers.

2. Steer new customers toward perishable categories in their first order

First orders anchored in produce or dairy correlate with meaningfully higher repeat rates (~54%) than shelf-stable-anchored first orders (~44-48%). Recommend testing merchandising nudges for new/early customers — e.g. featuring produce/dairy items in first-order promotions or onboarding flows — as a low-cost lever to shift more customers into the higher-retention category mix. This is a correlational finding, so should be validated with a controlled test before broader rollout.

3. Focus retention investment around the order 4-10 window

The steepest retention decay happens early (order 4 to order 10: 95.9% down to 52.5%), while the curve flattens at higher order counts. Retention spend is likely most efficient when concentrated on customers in this order-4-to-10 window, rather than spread evenly across the full customer lifecycle.

Tools & Deliverables

SQL: PostgreSQL cohort views — vw_retention_curve, vw_first_order_dept_retention, vw_basket_size_by_sequence, vw_days_since_prior_by_group. **Python:** pandas-based data loading, user-level sampling, and validation (referential integrity, sequence continuity, null checks). **Dashboard:** Power BI, 5-page report (Summary + 4 detail pages) with page navigation. **Sample:** 25,000 users, ~410,000 orders, ~4.1M order line items.